

ELEC 3EY4 - Final Prep → Technically Midterm 2

- 1 pg double sided cheat sheet & any calculator allowed
- no coding questions
-

- 10% → Graduate attributes from course outline

COURSE OBJECTIVES

By the end of this course, students should be able to:

- Understand and use basic commands in the Linux operating system
- Understand the principles of the Robot Operating System (ROS) and use it to develop and implement a real-time control system on Nvidia Jetson Nano AI embedded computer
- Analyze the magnetic and electrical characteristics of electric motor drives
- Use reference frame transformations to develop a model of a permanent magnet synchronous motor
- Understand the basic principles of sensing, planning, and control in self-driving vehicles
- Use coordinate frames and rigid transformations for expressing configuration and movement of rigid bodies in two and three-dimensional space
- Understand the principles of localization and mapping in mobile robotic systems
- Understand and develop feedback controllers for autonomous driving
- Understand principals of Quadratic Optimization and use it for developing an autonomous driving algorithm

Outcomes	Indicators	Measurement Methods(s)
Actively contributes to the planning and execution of a team project.	6.1	Lab assignments and reports
Manages interpersonal relationships, taking leadership responsibilities as needed.	6.2	Lab assignments and reports
Demonstrates comprehension of technical and non-technical instructions and questions.	7.1	Lab assignments and reports
Composes an effective written document for the intended audience.	7.2	Lab reports
Composes and delivers an effective oral presentation for the intended audience.	7.3	Lab and exam questions
Describes the duty of a Professional Engineer to the public, client, employer, and the profession.	8.1	Exam questions
Evaluates the social impact of engineering activities, including health, safety, legal, cultural, and other relevant factors, and identifies uncertainties in decisions.	9.2	Lab assignments and reports
Plans and effectively manages a project's time, resources, and scope, following business practices as appropriate.	11.2	Lab assignments and reports
Identifies, characterizes, assesses, and manages risks to project success.	11.3	Lab assignments and reports
Reflects on one's educational needs and opportunities for growth.	12.1	Lab and exam questions
Seeks and acquires appropriate external information as required, including showing awareness of sources of information and ability to critically evaluate them.	12.2	Lab and exam questions

Vehicle Kinematics

$$\dot{x} = v_s \cos \theta \rightarrow x_{i+1} = x_i + \Delta t \cdot \dot{x}_i$$

$$\dot{y} = v_s \sin \theta \rightarrow y_{i+1} = y_i + \Delta t \cdot \dot{y}$$

$$\dot{\theta} = \frac{v_s}{l} \tan \delta \rightarrow \theta_{i+1} = \theta_i + \Delta t \cdot \dot{\theta}_i$$

- ## Coordinate Frames

A diagram of a LEGO Mindstorms EV3 robot with a 3D coordinate system (x, y, z) centered on its top. The robot is equipped with a battery, a motor, and two large black wheels. To the right of the robot, there are two transformation matrices and a rotation formula:

$$R_{\text{map} \leftarrow \text{odom}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{\text{odom} \leftarrow \text{map}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\phi = -\frac{\pi}{2}, \quad \psi = -\frac{\pi}{2}$$

Below the matrices, there are three curved arrows representing rotations: a red arrow for ϕ , a green arrow for ψ , and a blue arrow for the yaw rotation.

Spatial Description of Rigid Bodies & Coord Frame

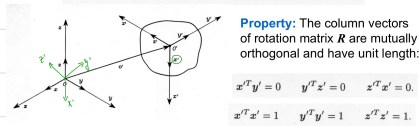
- Position** of a point O' on the body:
- $$o' = o'_x x + o'_y y + o'_z z,$$
- $$o' = \begin{bmatrix} o'_x \\ o'_y \\ o'_z \end{bmatrix}$$

-
- The parameters x', y', z' are the projection of u along the x, y and z directions, that is

$$z'_x = \langle \underline{z}', \underline{x} \rangle, \quad z'_y = \langle \underline{z}', \underline{y} \rangle, \quad z'_z = \langle \underline{z}', \underline{z} \rangle$$

- Rotation Matrix R :

$$R = \begin{bmatrix} x' & y' & z' \end{bmatrix} = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix} = \begin{bmatrix} x'^T x & y'^T x & z'^T x \\ x'^T y & y'^T y & z'^T y \\ x'^T z & y'^T z & z'^T z \end{bmatrix}$$


$$\mathbf{x}'^T \mathbf{y}' = 0 \quad \mathbf{y}'^T \mathbf{z}' = 0 \quad \mathbf{z}'^T \mathbf{x}' = 0.$$

$$\mathbf{x}'^T \mathbf{x}' = 1 \quad \mathbf{y}'^T \mathbf{y}' = 1 \quad \mathbf{z}'^T \mathbf{z}' = 1.$$

Elementary Rotations: Consider the body frame $O^*-x^*y^*z^*$ is obtained by rotating the reference frame $O-xyz$ by angle α wrt. the z axis (Figure). Then:

$$x' = \begin{bmatrix} \cos \alpha \\ \sin \alpha \\ 0 \end{bmatrix} \quad y' = \begin{bmatrix} -\sin \alpha \\ \cos \alpha \\ 0 \end{bmatrix} \quad z' = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \Rightarrow \quad R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B(x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \dots & \dots \end{bmatrix}$$

$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

In general, $R_k(-v) = R_k^{-1}(v) = R_k^T(v)$, $k = x, y, z$

Example 1: $O-x'y'z'$ is $O-xyz$ rotated by angle α about z axis (Figure 1):

$$p = Rp'$$

$$R_x(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p' = \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix}$$

$$p' = \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix}$$

$$p_x = p'_x \cos \alpha - p'_y \sin \alpha$$

$$p_y = p'_x \sin \alpha + p'_y \cos \alpha$$

$$p_i = p'_i.$$

Example 2 (Rotation of Vectors): Consider vector $\mathbf{p}$ in the xy plane that is obtained by rotating a vector $\mathbf{p}'$ about the z axis of the reference frame O - xyz by angle α (Fig. 2). In this case, from Figure 2

In this case, from

$$p_x' = \|\underline{p}\| \cos(\beta)$$

$$p_x = \| \underline{p} \| \cos (\alpha + \beta)$$

$$p_y = \| \underline{p} \| \sin (\alpha + \beta)$$

$$\|p\| = \|p'\|$$

$$p_x = p'_x \cos \alpha - p'_y \sin \alpha$$

$$p_y = p'_x \sin \alpha + p'_y \cos \alpha$$

$$p_z = p'_z.$$

$$p = R_x(\alpha)p',$$

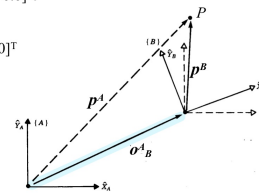
$$\mathbf{R}_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{o}^A_B = [10 \ 5 \ 0]^T$$

$$A^A_B = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 10.0 \\ 0.500 & 0.866 & 0.000 & 5.0 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{p}^B = [3.0, 7.0, 0.0, 1]^T$$

$$p^A = A^A_B p^B \quad \Rightarrow \quad p^A = [9.1, 12.6, 0.0, 1]^T$$

$$\mathbf{p}^A = [9.1, 12.6, 0.0]^T$$



$$\rho^A = 0^A_B + R^A_B \rho^B$$

$$= A^A_B \rho^B$$

ELECENG 3EY4: Electrical Systems Integration Project
Final Exam
Instructor: Shahin Sirouspour

CLASS: DAY

DURATION OF EXAMINATION: 120 min.

DATE: APRIL 14, 2023

THIS EXAMINATION PAPER INCLUDES SIX (6) PAGES AND FOUR (4) QUESTIONS. YOU CAN EARN A TOTAL OF 100 MARKS FROM THIS TEST. BRING ANY DISCREPANCY TO THE ATTENTION OF YOUR INVIGILATOR.

Special Instructions:

- (a) The exam is closed book. You are allowed to use a double-sided letter size crib sheet.
- (b) You are permitted to use any calculator.
- (c) You must answer the questions in the answer booklet provided to you.
- (d) You must return this exam paper along with the answer booklet.

START OF EXAM QUESTIONS

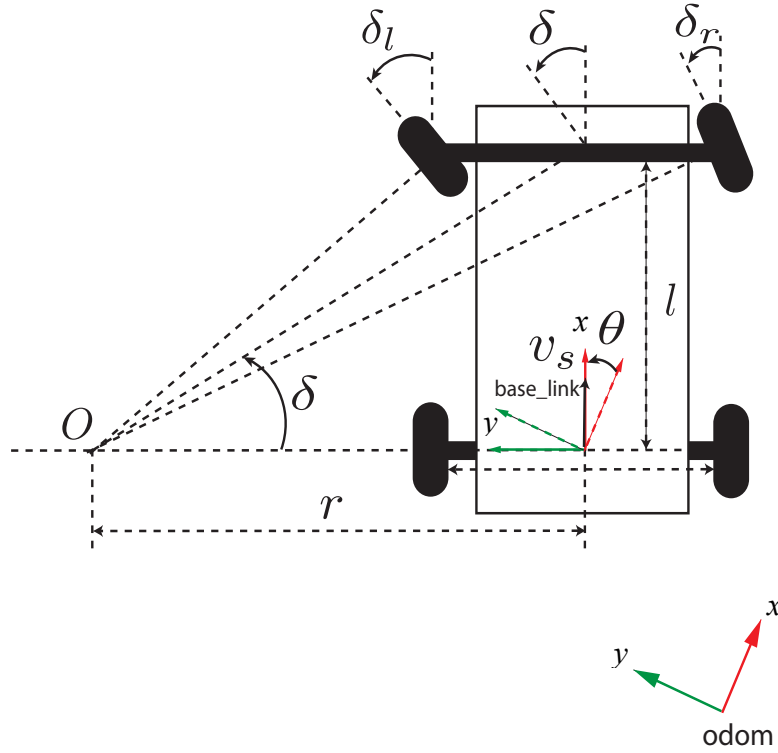


Figure 1: Vehicle schematics with odometry (odom) and vehicle body frames

QUESTION #1 (25 marks)

Consider the motion model of MacAEV in Fig. 1:

$$X = [x \quad y \quad \theta]^T$$

$$\dot{x} = v_s \cos \theta$$

$$\dot{y} = v_s \sin \theta$$

$$\dot{\theta} = \frac{v_s}{l} \tan \delta$$

Here x, y are the position and θ is the yaw orientation of the frame “base_link” with respect to the frame “odom”; v_s is the vehicle forward velocity and δ is the vehicle steering angle.

Assume at $t = 1$, $X(1) = [1.0 \quad -2.0 \quad \frac{\pi}{4}]^T$. The user velocity and steering angle commands are given by:

$$v_s(t) = \begin{cases} 2 & 1 \leq t < 2 \\ 1 & 2 \leq t \leq 3 \end{cases} \quad \theta(t) = \begin{cases} 0.1 & 1 \leq t < 2 \\ -0.1 & 2 \leq t \leq 3 \end{cases}$$

- a) Find the vehicle configuration in space at $t = 3$, $X(3) = ?$ using wheel-based odometry. (8 marks)

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- Find the vehicle configuration in space at $t = 3$, $X(3) = ?$ using wheel-based odometry. (8 marks)
- Referring to Figs. 2&3, find the rotation matrix R_l^b , and the translation vector t_l^b relating the frame "base_link" to the frame "laser". (8 marks)
- If at $t = 1$, the LiDAR reports an obstacle point with the range and angle $(r, \alpha) = (4, \frac{\pi}{4})$, what would be the reported range and angle of the same obstacle point at $t = 3$? (Note that the angle is measured with respect to the x -axis of the LiDAR frame). (9 marks)

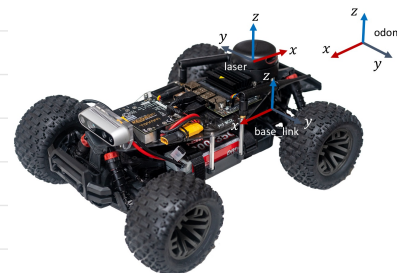
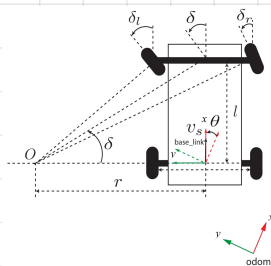


Figure 2: MacAEV Frame Assignment

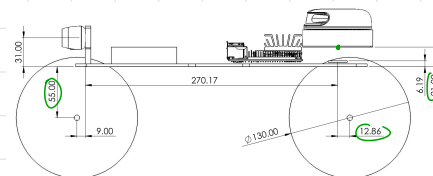


Figure 3: MacAEV Assembly Diagram

a) $t = 1: X(1) = [1 \quad -2 \quad \frac{\pi}{4}]^T, v_s(1) = 2, \delta(1) = 0.1$

$$\begin{aligned} x_2 &= x_1 + v_s \cos \theta_1 & y_2 &= y_1 + v_s \sin \theta_1 & \theta_2 &= \theta_1 + \frac{v_s}{l} \tan \delta \\ &= 1 + 2 \cos(\frac{\pi}{4}) & &= -2 + 2 \sin(\frac{\pi}{4}) & &= \frac{\pi}{4} + \frac{2}{0.27017} \tan(0.1) \\ &= 2.41 & &= -0.586 & &= 1.53 \end{aligned}$$

$$t = 2: X(2) = [x_2 \quad y_2 \quad \theta_2]^T, v_s(2) = 1, \delta(2) = -0.1$$

$$= [2.41 \quad -0.586 \quad 1.53]^T$$

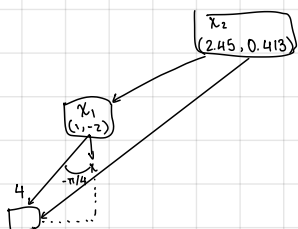
$$\begin{aligned} x_3 &= x_2 + v_s \cos \theta_2 & y_3 &= y_2 + v_s \sin \theta_2 & \theta_3 &= \theta_2 + \frac{v_s}{l} \tan \delta \\ &= 2.41 + 1 \cos(1.53) & &= -0.586 + 1 \sin(1.53) & &= 1.53 + \frac{1}{0.27017} \tan(-0.1) \\ &= 2.45 & &= 0.413 & &= -1.16 \end{aligned}$$

$$t = 3: X(3) = [2.45 \quad 0.413 \quad 1.16], v_s = 1, \delta = -0.1$$

b) $R_l^b = ?$ & $t_l^b = ?$

$$R_l^b = \begin{bmatrix} z & y & x \\ 0 & 1 & 0 \end{bmatrix} \quad t_l^b = \begin{bmatrix} x & y & z \\ 12.86 & 0 & 71.0 \end{bmatrix}$$

c) $X(1) = [1 \quad -2 \quad \frac{\pi}{4}] \rightarrow (r_1, \alpha_1) = (4, \frac{\pi}{4})$
 $X(3) = [2.45 \quad 0.413 \quad 1.16] \rightarrow (r_3, \alpha_3) = ?$



- b) Referring to Figs. 2&3, find the rotation matrix R_l^b , and the translation vector t_l^b relating the frame “base_link” to the frame “laser”. (8 marks)
- c) If at $t = 1$, the LiDAR reports an obstacle point with the range and angle $(r, \alpha) = \left(4, \frac{-\pi}{4}\right)$, what would be the reported range and angle of the same obstacle point at $t = 3$? (Note that the angle is measured with respect to the x -axis of the LiDAR frame). (9 marks)

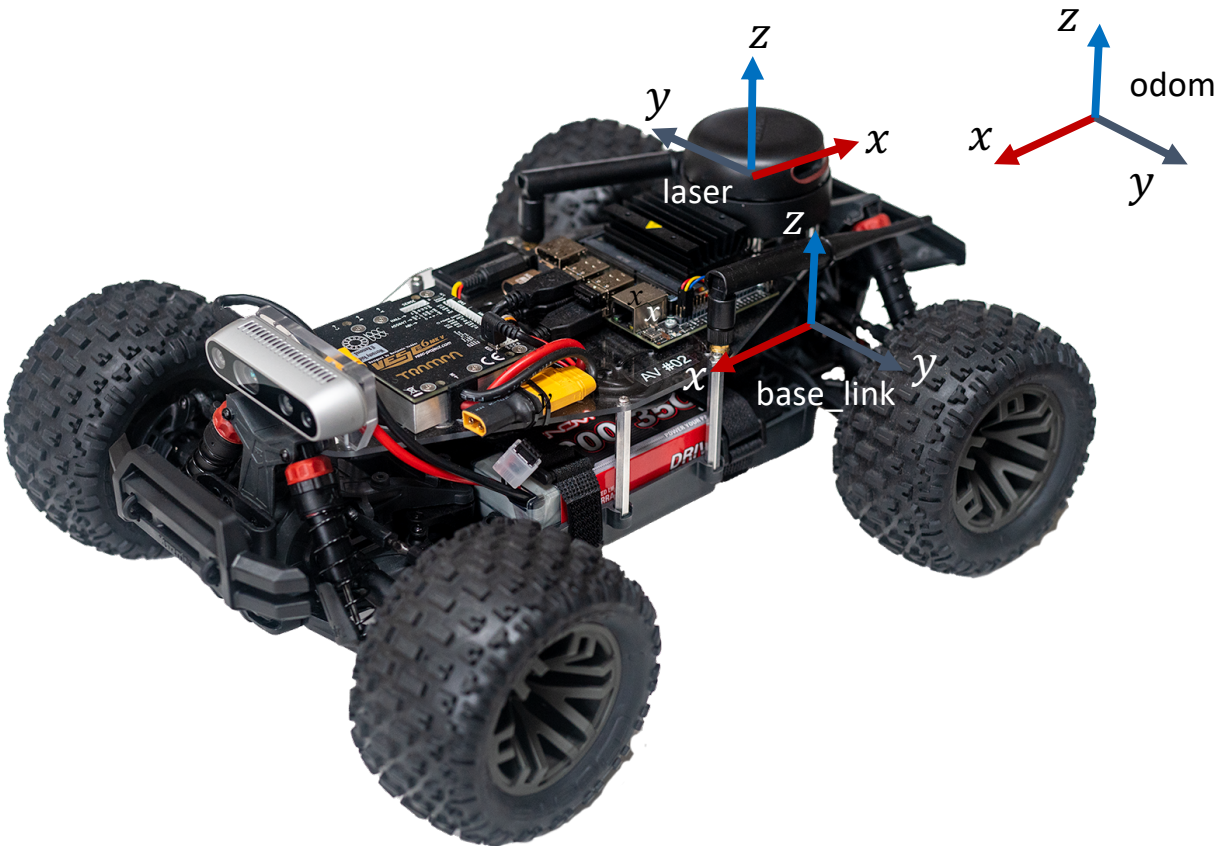


Figure 2: MacAEV Frame Assignment

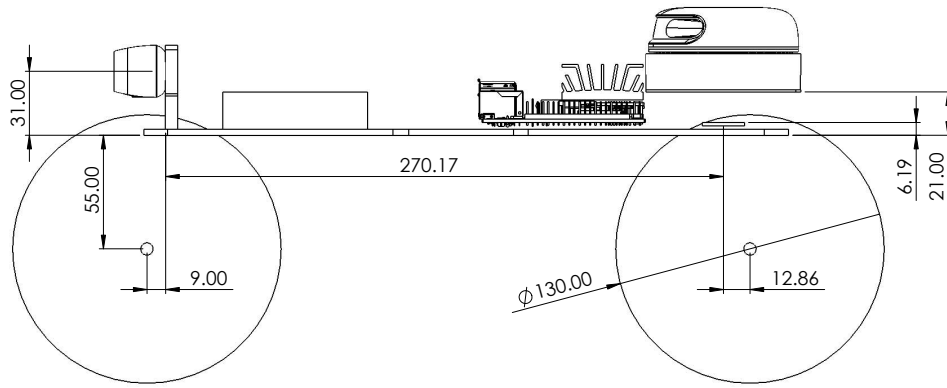


Figure 3: MacAEV Assembly Diagram

QUESTION #2 (25 marks)

The Laser Scan Matching algorithm can be used to determine vehicle motion between the times of two consecutive scans of LiDAR data. Assume that you are given two sets of LiDAR data points $\{p_i\}$ at time $t = 0$ and $\{q_j\}$ at time $t = 1$, both expressed in a LiDAR-attached frame. Your goal is to find the unknown rotation matrix $R_1^0(\theta)$, and translation vector t_1^0 between the LiDAR frames at $t = 0$ and $t = 1$. Note that the rotation matrix here is function of only one variable, i.e., the yaw angle θ . Explain, with supporting mathematical formulation as needed, the process by which these unknown values can be found in the Scan Matching algorithm.

QUESTION #3 (25 marks)

Referring to Fig. 4, the problem of finding a separating virtual barrier between the origin of a vehicle-attached frame and a set of LiDAR returns $\{p_i\}$ can be formulated as a *Quadratic Optimization*:

$$\begin{aligned} \min_w \quad & w^T w \\ \text{s.t.} \quad & w^T p_i + 1 \leq 0 \quad i = 1, \dots, n \end{aligned}$$

where $w \in \mathbb{R}^2$ is the vector that characterizes the virtual barrier line.

- Explain how this optimization problem formulation is derived. (10 marks)
- Show how this optimization problem can be solved using an *active set* approach and Lagrangian formulation for Quadratic Optimization with *equality* constraints. Present any relevant mathematical formulation. (15 marks)

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$$R_1^0 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

t_2

t_1

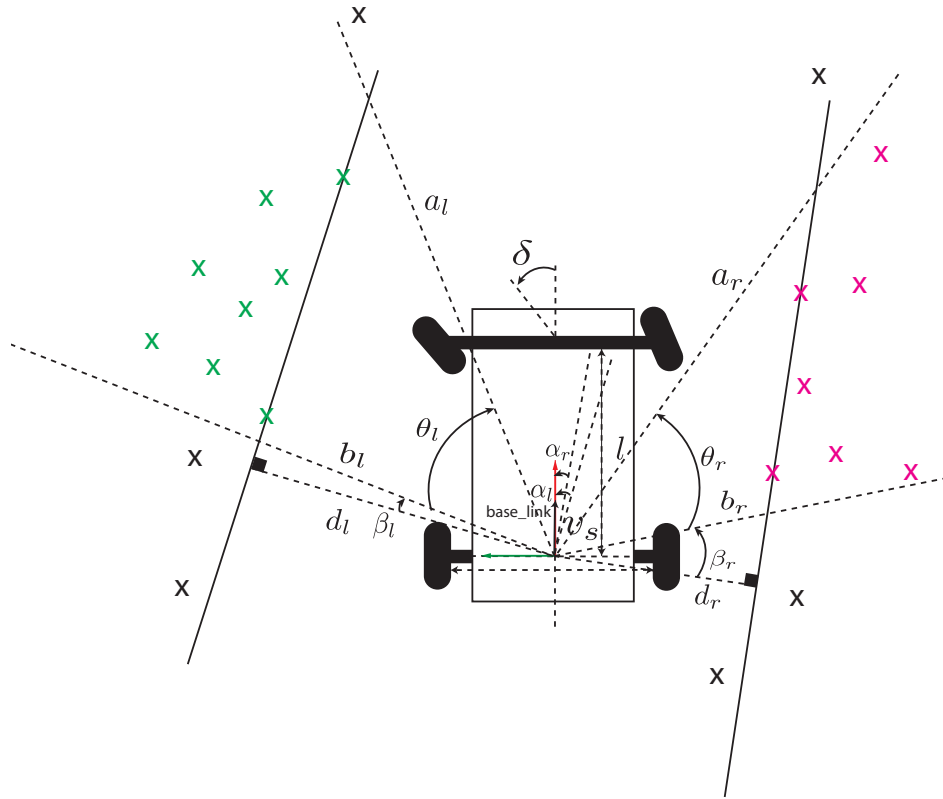


Figure 4: Finding a separating virtual barrier between the vehicle and set of LiDAR returns.

QUESTION #4 (25 marks)

This question is concerned with the multi-view vision system in Fig. 5.

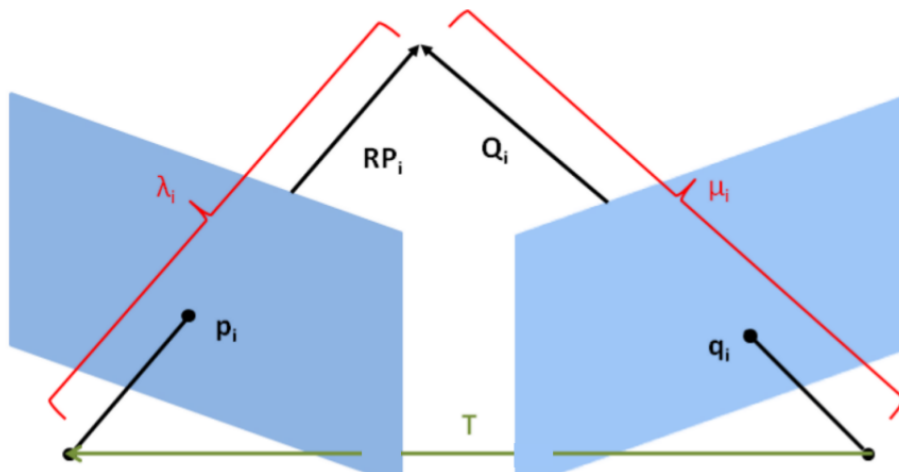


Figure 5: Multi-view vision system

In this system, two cameras capture images of a single point in 3D space. Here P_i, Q_i are the coordinates of the point in the left and right camera frames, respectively; p_i, q_i are the coordinates of the projections of the point in the image planes of the left and right camera, expressed in their respective frames. Moreover, $\lambda_i, \mu_i > 0$ are two scalars. Note that:

$$P_i = \lambda_i p_i, \quad Q_i = \mu_i q_i$$

$$p_i = \begin{bmatrix} p_i^x \\ p_i^y \\ f \end{bmatrix}, \quad q_i = \begin{bmatrix} q_i^x \\ q_i^y \\ f \end{bmatrix}$$

and f is the focal length. Finally, $R \in \mathbb{R}^{3 \times 3}, T \in \mathbb{R}^{3 \times 1}$ are the rotation and translation of the left camera frame with respect to the right camera frame.

- Write the equation that relates the coordinates of the projections of the point on the two image planes, i.e., p_i, q_i . (6 marks)
- For a given point in the left image plane p_i , find all possible corresponding points in the right image plane, q_i . What characterizes these points? (6 marks)
- Show how the depth ambiguity in single-view vision can be resolved in the multi-view vision. You must explain how corresponding points in the two images may be found, and then how that 3D coordinates of each image point are computed. (13 marks)

THE END